



Packet Tracer VPN Configuration

☑ Completed	☑
⚙ Status	To do: Lab Work

Introduction

This activity covers the configuration of an IPsec VPN. IPsec is an open standard used for configuring secure Virtual Private Networks over the internet through IPsec tunnels, which provide secure, encrypted connections. These tunnels are commonly used for site-to-site VPNs to securely connect networks, which this activity implements. In summary, site-to-site VPNs allow remote site networks, not in the same network, to connect with the network and be seen as a local network segment.

Internet Key Exchange (IKE), also called Internet Security Association and Key Management Protocol (ISAKMP), is the negotiation protocol that lets two hosts agree on how to build an IPsec Security Association (SA), which defines the security parameters for the tunnel. ISAKMP separates negotiation into two phases, in which Phase 1 creates the first tunnel, which protects later ISAKMP negotiation messages, and Phase 2 creates the tunnel that protects the data.

Creating an ISAKMP policy includes the following:

- An authentication method to ensure the identity of the peers.
- An encryption method to protect the data and ensure privacy.
- A hashed Message Authentication Code (HMAC) method to ensure the identity of the sender, and to ensure that the message has not been tampered with.
- A Diffie-Hellman group to determine the strength of the encryption-key-determination algorithm to derive the encryption and hash keys.

- Limit the time the security appliance uses as an encryption key before replacing it.

The main purpose of IKE Phase 2 is to negotiate the IP security parameters that will be used to secure the IPsec Tunnel. Security Associations (SAs) are negotiated by the IKE process on behalf of IPsec. SAs define the protocols and algorithms to apply to sensitive data and also specify the keying material the hosts will use. IPsec SAs control the actual transmission of user traffic. SAs are unidirectional but are generally established in pairs, inbound and outbound. The hosts negotiate the settings to use for each SA, which consists of the following:

- Transform sets: a combination of security protocols and algorithms that define how the appliance protects data, which should be the same for both peers.
- Crypto maps: define the IPsec policy to be negotiated in the IPsec SA, which includes the access list, peer identification, local address for the IPsec traffic, and up to six transform sets.
- Access lists
- Tunnel groups
- Prefragmentation Policies

Cisco ACLs are single/multiple permit or deny statements that filter inbound or outbound packets on a network interface. Unlike standard ACLs, which are implemented close to the destination, extended ACLs are implemented close to the source and are granular specific specifying the source and destination addresses, protocols, and port numbers.

Standard Numbered ACLs, which range from 1- 99 and 1300 – 1999, specify permit or deny traffic from only a source address with optional wildcard masks, which are used for filtering subnet ranges. With extended numbered ACLs, which range from 100 – 199 and 2000 – 2699, support multiple permit and deny statements with source and/or destination IP addresses. There is an implicit hidden deny all statement clause; therefore, you must include <permit ip any any> to effectively permit packets that do not match any of the previous clauses within an ACL.

Objective

Part 1: Enable Security Features on R1 & R3

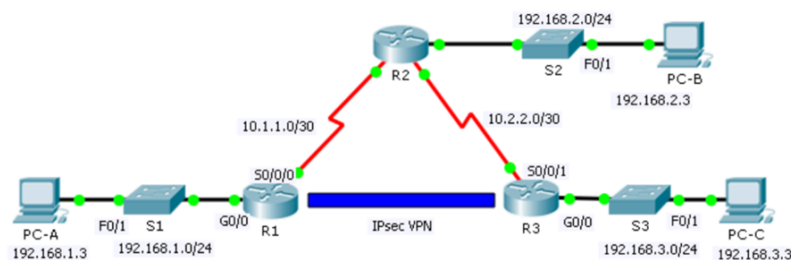
Part 2: Configure IPsec Parameters on R1 & R3

Part 3: Verify the IPsec VPN

Scenario

The already configured network topology, as shown below, shows three routers.

Topology



With the following configurations

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	10.1.1.2	255.255.255.252	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A
	S0/0/0	10.1.1.1	255.255.255.252	N/A
R3	G0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/1	10.2.2.2	255.255.255.252	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1

The task was to configure R1 and R3 with the following parameters to support a site-to-site VPN when traffic flows between their respective LANs. The IPsec VPN tunnel is from R1 to R3 via R2, which acts as a pass-through and has no knowledge of the VPN.

ISAKMP Phase 1 Policy Parameters

Parameters		R1	R3
Key distribution method	Manual or ISAKMP	ISAKMP	ISAKMP
Encryption algorithm	DES , 3DES, or AES	AES	AES
Hash algorithm	MD5 or SHA-1	SHA-1	SHA-1
Authentication method	Pre-shared keys or RSA	pre-share	pre-share
Key exchange	DH Group 1 , 2, or 5	DH 2	DH 2
IKE SA Lifetime	86400 seconds or less	86400	86400
ISAKMP Key		cisco	cisco

Bolded parameters are defaults. Other parameters need to be explicitly configured.

IPsec Phase 2 Policy Parameters

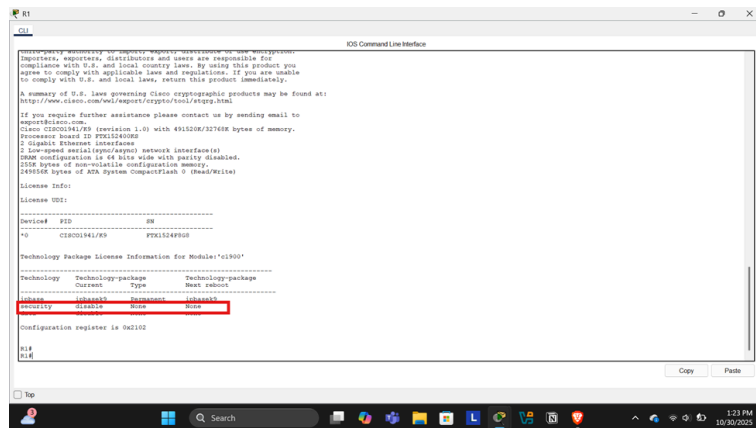
Parameters	R1	R3
Transform Set	VPN-SET	VPN-SET
Peer Hostname	R3	R1
Peer IP Address	10.2.2.2	10.1.1.2
Network to be encrypted	192.168.1.0/24	192.168.3.0/24
Crypto Map name	VPN-MAP	VPN-MAP
SA Establishment	ipsec-isakmp	ipsec-isakmp

Part 1: Enable Security Features on R1 & R3

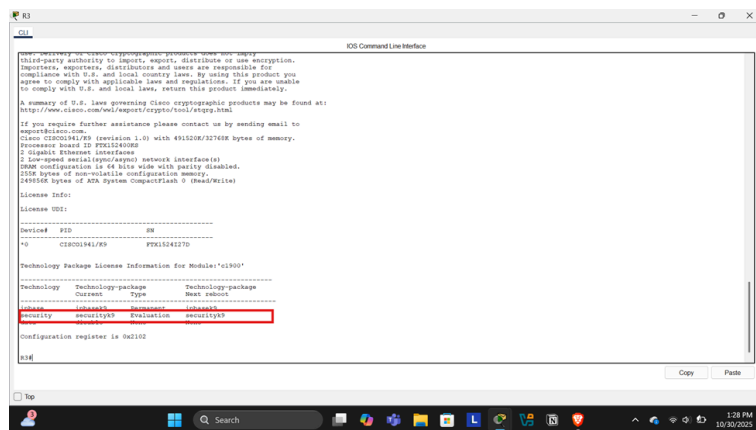
Step 1: Activate the securityk9 module

SecurityK9 is a security technology package license that enables full-featured, high-level encryption for features such as IPsec and others, including SSL/SSH and VPNs on Cisco Routers. The package must be enabled on the devices, in this case, the routers.

1. Used the <show version> command to verify if the security package exists and is active. As highlighted below and reflected on both router R1 & R3, no security packages were installed or active.
 - a. R1 Configuration

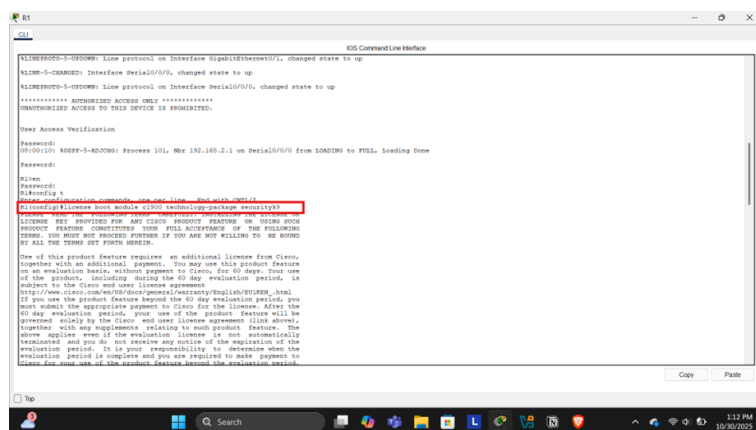


b. R3 is already configured with the security package; therefore, the next steps (b) and (c) are applied to devices that do not have the security package installed.



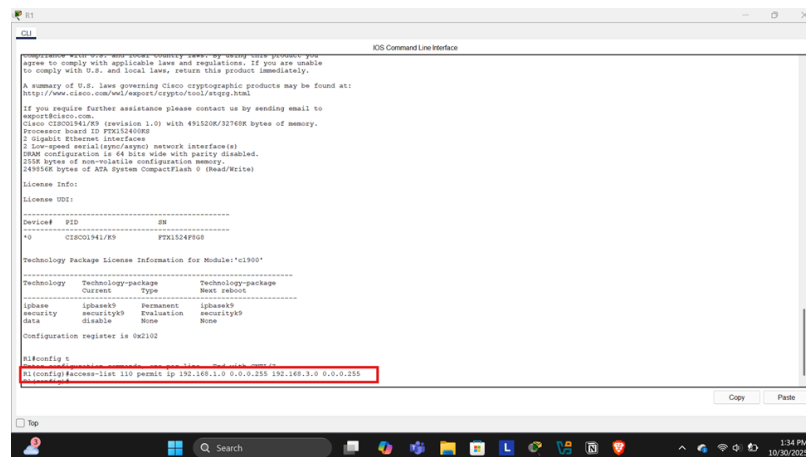
2. Enabled the security technology package on both R1 and R3 with the same command process, accepted the end-user license agreement, saved the configurations to ensure the configurations are there when restarting the machine, and reloaded the router to enable the security license.

a. R1 Configuration



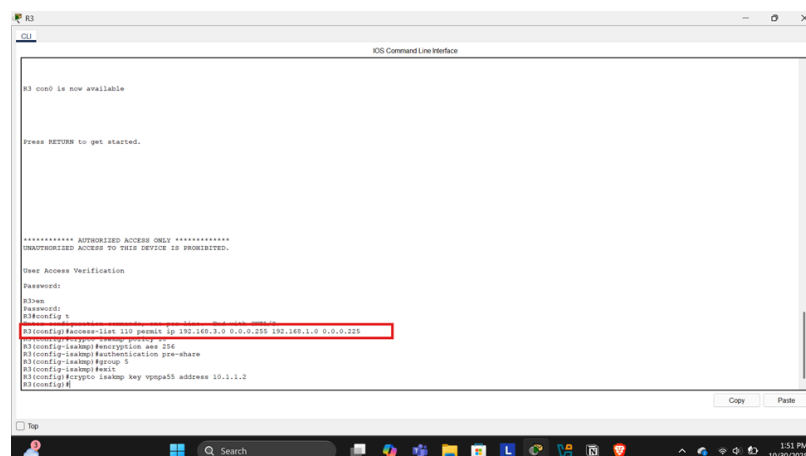
Configured ACL 110 to identify the traffic from the LAN in R1 to the LAN in R3 as interesting. This interesting traffic will trigger the IPsec VPN to be implemented when there is traffic between the R1 to R3 LANs.

1. R1 Configuration



```
CU>  
Copyright 1986-2008 Cisco Systems, Inc. All rights reserved. Cisco Confidential  
KIOS Command Line Interface  
*****  
If you require further assistance please contact us by sending email to support@cisco.com.  
Cisco IOS XE 3.14.1 (revision 1.0) with 491520K/32768K bytes of memory.  
Processor board ID FTX1524F0G0  
2 Gigabit Ethernet interfaces  
2 Serial Async Serial Async network interface(s)  
DRAM configuration is 64 bits wide with parity disabled.  
256K bytes of non-volatile configuration memory.  
2459568 bytes of ATA System CompactFlash 0 (Read/Write)  
License Info:  
License UDI:  
-----  
Device# PID SR  
-----  
*0 CISC01941/R3 FTX1524F0G0  
Technology Package License Information for Module: "c1900"  
-----  
Technology Technology-package Technology-package  
Current Type Best subord  
-----  
ipbase ipbasek9 Permanent ipbasek9  
security securityk9 Evaluation securityk9  
data datak9 None  
Configuration register is 0x2102  
R1#  
R1(config)#  
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255  
R1(config)#
```

2. R3 Configuration



```
CU>  
R3 com0 is now available  
Press RETURN to get started.  
***** AUTHORIZED ACCESS ONLY *****  
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.  
Enter Access Verification  
Password:  
R3>  
Password:  
R3#  
R3(config)#  
R3(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255  
R3(config)#  
R3(config)#crypto isakmp key vpnpa55 address 10.1.1.2  
R3(config)#
```

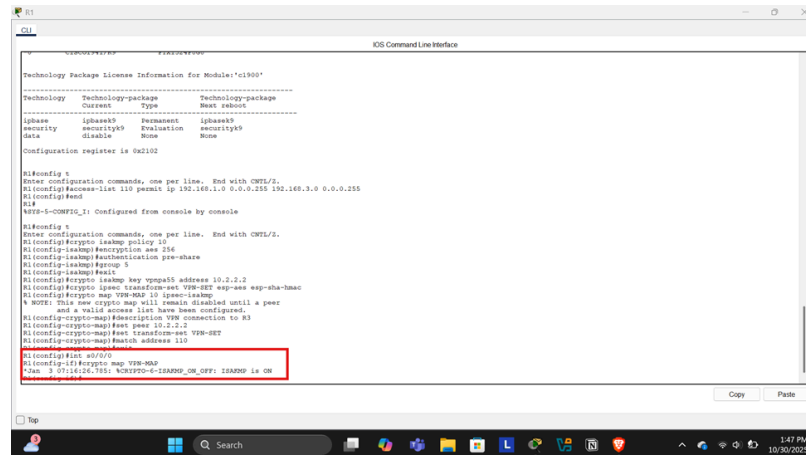
Step 2: Configure the IKE Phase 1 ISAKMP policy

Configured the crypto ISAKMP policy, with a priority of 10, properties on R1 and R3, along with the shared crypto key vpnpa55, with the configuration highlighted in the ISAKMP Phase 1 table. The encryption method in which AES of key length 256 bits was used, the key exchange method in which pre-shared keys were used, which is not scalable but suitable for smaller networks, as this, and the DH method of group 5, which supports AES, were configured.

1. R1 Configuration

appliance to evaluate all the traffic against the crypto map set and to use the specified policy during connection or SA negotiation.

1. R1 Configuration



```
RT
CU
KIOS Command Line Interface

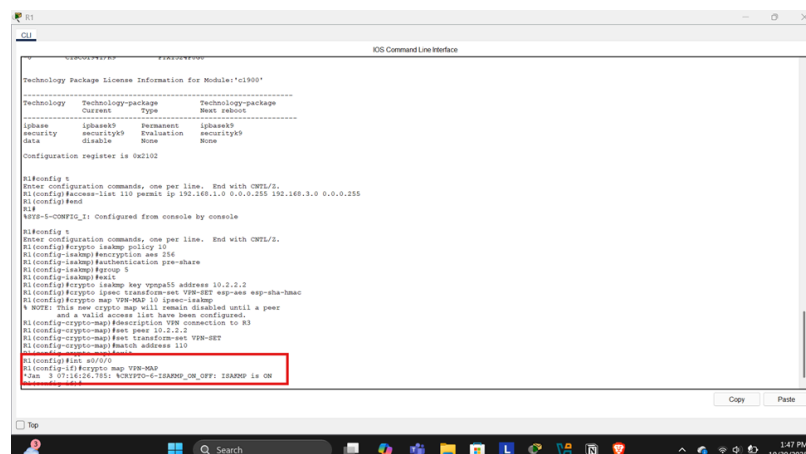
Technology Package License Information for Module:"c1900"
-----
Technology    Technology-package    Technology-package
Current      Type                Best subopt
-----
ipbase       ipbasek9             Permanent          ipbasek9
security     security9             Evaluation          security9
data         disable              None               None

Configuration register is 0x2102

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
R1(config)#end
R1#
R1#S1-S-CONFID_I: Configured from console by console

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key vpnga5 address 10.2.2.2
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
* NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R1(config)#crypto map description VPN connection to R3
R1(config)#crypto-map peer 10.2.2.2
R1(config)#crypto-map peer transform-set VPN-SET
R1(config)#crypto-map match address 110
R1(config)#exit
R1(config)#crypto map VPN-MAP
*Jan 3 07:16:24.755: NCPPTD-6-12ARPDP_ON_OFF: 12ARPDP is ON
```

2. R3 Configuration



```
RT
CU
KIOS Command Line Interface

Technology Package License Information for Module:"c1900"
-----
Technology    Technology-package    Technology-package
Current      Type                Best subopt
-----
ipbase       ipbasek9             Permanent          ipbasek9
security     security9             Evaluation          security9
data         disable              None               None

Configuration register is 0x2102

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
R3(config)#end
R3#
R3#S1-S-CONFID_I: Configured from console by console

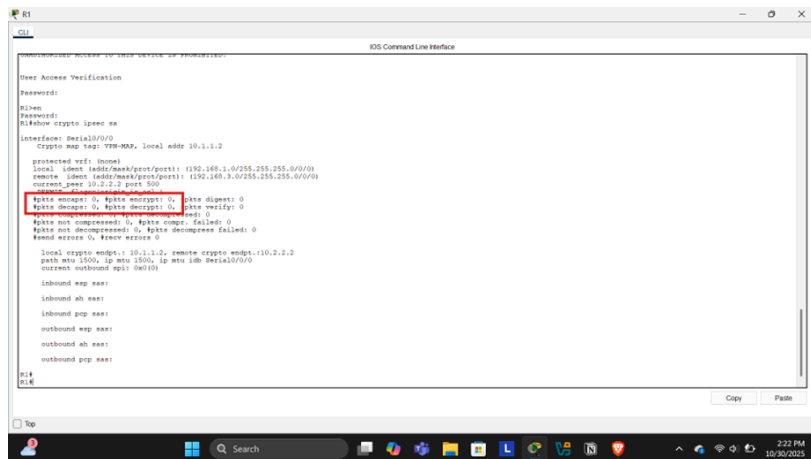
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#group 5
R3(config-isakmp)#exit
R3(config)#crypto isakmp key vpnga5 address 10.2.2.2
R3(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R3(config)#crypto map VPN-MAP 10 ipsec-isakmp
* NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R3(config)#crypto map description VPN connection to R1
R3(config)#crypto-map peer 10.2.2.2
R3(config)#crypto-map peer transform-set VPN-SET
R3(config)#crypto-map match address 110
R3(config)#exit
R3(config)#crypto map VPN-MAP
*Jan 3 07:16:24.755: NCPPTD-6-12ARPDP_ON_OFF: 12ARPDP is ON
```

Part 3: Verify the IPsec VPN

Step 1: Verify the tunnel before interesting traffic

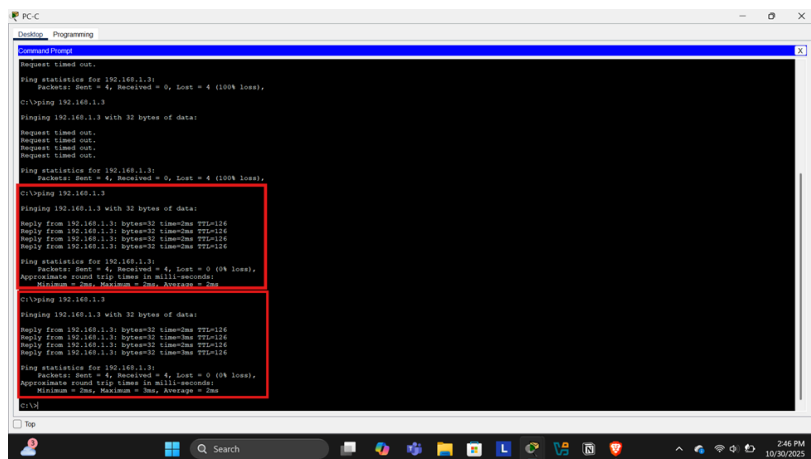
Used the <show crypto ipsec sa> command on R1 to examine the state of the packets before traffic is passed through the tunnel.

1. R1



Step 2: Create interesting traffic

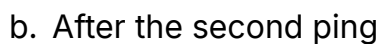
1. Ping PC-C from PC-A



Step 3: Verify the tunnel after interesting traffic

Used the <show crypto ipsec sa> command on R1 to examine the state of the packets after traffic passes through the tunnel.

1. R1
 - a. After the first ping



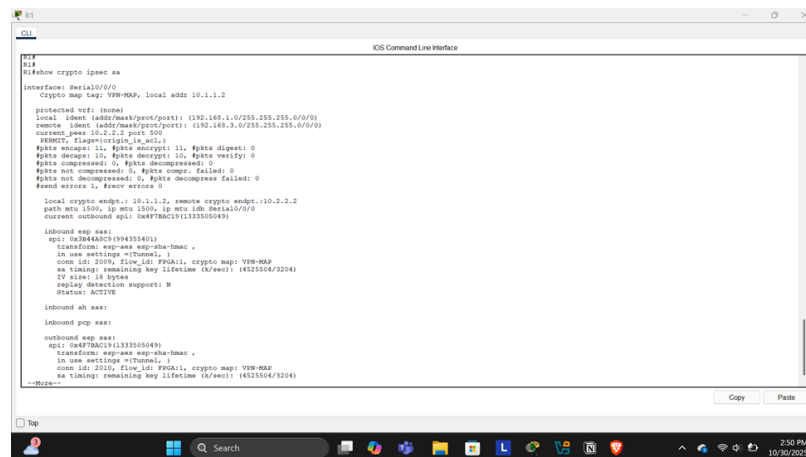
Pinged PC-A from PC-B, where traffic does not go through the IPsec VPN configured.

1. Ping PC-A from PC-B



Used the <show crypto ipsec sa> command on R1 to examine the state of the packets after traffic did not pass through the tunnel. As highlighted below, there was no change in the number of packets.

1. R1



```
R1
R1#show crypto ipsec sa
Interface: Serial0/0/0
Crypto map tag: VPM-001, local addr 10.1.1.2

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
current peer 10.2.2.2 port 500
PERMIT: flags=(origin, is_acl)
  Rpkts encap: 11, Rpkts encrpt: 11, Rpkts digest: 0
  Rpkts decap: 10, Rpkts decrpt: 10, Rpkts verify: 0
  Rpkts compr: 0, Rpkts decompr: 0
  Rpkts not compr: 0, Rpkts compr. failed: 0
  Rpkts not decompr: 0, Rpkts decompr. failed: 0
  Rsend errors: 1, Rrcv errors: 0

local crypto endpt.: 10.1.1.2, remote crypto endpt.: 10.2.2.2
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
current outbound spi: 0x8F78AC19(1333505049)

inbound esp sa:
  spi: 0x3646AC5(994355401)
  transform: esp-aes esp-sha-hmac
  in use settings =Tunnel, 1
  conn id: 2009, flow_id: P0A1, crypto map: VPM-001
  sa timing: remaining key lifetime (s/sec): (4325504/3204)
  IV size: 16 bytes
  replay detection support: N
  Status: ACTIVE

inbound ah sa:

inbound pcp sa:

outbound esp sa:
  spi: 0x8F78AC19(1333505049)
  transform: esp-aes esp-sha-hmac
  in use settings =Tunnel, 1
  conn id: 2009, flow_id: P0A1, crypto map: VPM-001
  sa timing: remaining key lifetime (s/sec): (4325504/3204)
  --More--
```

Conclusion

This activity highlighted how to configure IPsec and ISAKMP standards to build Virtual Private Networks, which provide security through an encrypted tunnel. Given that the IPsec protocol secures layer 3 of the OSI model to encrypt and authenticate packets, ensuring security against vulnerable attacks such as routing attacks, DOS/DDOS, and IP spoofing.